Emma's Atlas of Numbers and Shapes

- A not-so-brief guide to all the math I know -

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2026-04-30

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Part I

Foundations

Chapter 1

An Introduction to Mathematical Reasoning

Chapter 2

Sets

Almost all of modern mathematics can be put in the context of the study of ***sets*** and their ***elements***. Generally, given a collection of objects $a, b, c, d, \ldots$, they can be taken together as a set, generally written in curly brackets and represented by a capital letter:

$$M := \{a, b, c, d, e, \ldots\}$$

However, an approach where any collection of mathematical objects forms a set quickly runs into trouble - famously, ***Russel's paradox*** asks:

> *"Let R be the set of all sets that don't contain themselves. Does R contain itself?"*

If R doesn't contain itself, then by definition it must contain itself, but if it contains itself, then it cannot contain itself! If $R \in R$ is true, then it must be false! We arrive at a classic paradox. The conclusion, then, is that the set of all sets that don't contain themselves cannot exist - we cannot define sets to be any collection of things defined by any property, and must instead establish ground rules about which sets exist and exactly how we can use sets we already know to construct new ones.

These ground rules are given in the form of ***axioms***, which are statements that mathematicians take for granted, which can then be used as a starting point to then derive "the rest of mathematics" from. The following chapters contain a brief and hopefully easily understandable introduction to the ZFC axiom system, which was developed in the early 20th century by Ernst Zermelo and Abraham Fraenkel and is now the most widely accepted axiomatization of mathematics.

2.1 Infinity: $\emptyset$ and $\mathbb{N}$

2.2 Extensionality

To say "b is an element of M", or equivalently "b contains M", we simply write:

$$b \in M$$

One of the most basic axioms of set theory is the ***Axiom of Extensionality***, which states:

Axiom 2.2.1: Axiom of Extensionality

Two sets A and B are equal if and only if they contain the same elements.

The Axiom of Extensionality establishes that sets are uniquely defined by the objects they contain and have no additional structure. In particular:

1. The order of elements in a set doesn't matter. $\{1, 2\}$ and $\{2, 1\}$ are the same set.
2. The number of times an element appears within the same set doesn't matter. $\{1\}, \{1, 1\}$, and $\{1, 1, 1\}$ are the same set.

Definition 2.2.2: Subsets

We call a set A a ***subset*** of a set B if and only if every element of A is contained in B. We write "A is a subset of B" as $A \subseteq B$.

The demand that "every element of A is contained in B" can be stated more formally as "$x \in A$ implies $x \in B$". Importantly, this means that every set is a subset of itself - we always have $A \subseteq A$.

This definition gives us another way of phrasing the Axiom of Extensionality - two sets are equal if and only if each is a subset of the other:

$$A = B \Leftrightarrow A \subset B \text{ and } B \subset A$$

This idea is incredibly important for writing out proofs in practice - if a problem asks you to establish that two sets A and B are the same, then the easiest approach is almost always to show that any element of A must be contained in B, and to then show that any arbitrary element of B must be contained in A.

The subset relation is frequently written as $A \subset B$. However, other authors reserve $A \subset B$ to mean "A is a subset of B that isn't equal to B" - we call such a set a ***proper subset of B***. I will therefore attempt to only use the symbols $\subseteq$ and $\subsetneq$, which clear up this ambiguity.

2.3 Specification

2.4 Pairing, Union, and Power Sets

2.5 Relations

2.6 Functions

2.7 Replacement

2.8 Regularity and Well-Foundedness

2.9 Choice

2.10 Cardinality

Chapter 3

Algebraic Structures

3.1 $(\mathbb{N}, +)$ is a Monoid

3.2 $(\mathbb{Z}, +, -)$ is a Group

3.3 $(\mathbb{Z}, +, -, \cdot)$ is a Ring

3.4 $(\mathbb{Q}, +, -, \cdot, ^{-1})$ is a Field

3.5 `Set` is a Category

Part II

Foundations of Analysis

Chapter 1

The Real Numbers $\mathbb{R}$

1.1 Ordered Fields

1.2 Dedekind Cuts

1.3 Uniqueness of $\mathbb{R}$

Chapter 2

The Concept of Distance

2.1 Metric Spaces

2.2 Convergence

2.3 Euler's Number e

2.4 Continuity

Chapter 3

Infinite Series

3.1 Achilles and the Turtle

3.2 Convergence Tests

Theorem 3.2.1: n-th Term Test

Let a_k be a sequence in $\mathbb{R}$ or $\mathbb{C}$ such that $\sum_{k=0}^{n} a_k$ converges. Then a_k converges to 0.

Proof. content... □

3.3 The Exponential Function

Chapter 4

Euclidean Geometry

4.1 The Complex Numbers $\mathbb{C}$

4.2 Circles, Angles and Trigonometry

Chapter 5

Function Spaces

Chapter 6

Differential Calculus in One Variable

Chapter 7

The Riemann-Stieltjes Integral

Part III

Linear Algebra

Chapter 1

Analytic Geometry

1.1 Arithmetic in n dimensions

We now focus our attention on n-dimensional euclidean space, which is modelled by the cartesian product $\mathbb{R}^n$:

$$\mathbb{R}^n = \left\{ \vec{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \;\middle|\; x_1, \ldots, x_n \in \mathbb{R} \right\} = \left\{ \vec{x} : \{1, \ldots, n\} \to \mathbb{R} \right\}$$

Elements of $\mathbb{R}^n$ are known as ***vectors***[1]. In the context of linear algebra, they are usually written vertically, as ***column vectors***.

[1] We will see in the next chapter that $\mathbb{R}^n$ is an example of a ***vector space***, including the case $n = 1$ (and $n = 0$, if you want to get a bit pathological). Therefore, formally speaking and only in certain contexts, regular old real numbers can be vectors too. However, when encountering the term "vector" in the wild, it is generally implied that $n \geq 2$.

Vectors simultaneously represent both ***positions*** and ***directions*** in space - in the same way that the number 5 is both the point that lies 5 units away from 0 on the real number line *and* the distance between the numbers 3 and 8 on that same number line, the vector $\binom{1}{2}$ represents both

- The point on the cartesian plane that can be reached by going one unit in the direction of one coordinate axis and then two units in the direction of the other axis, *and*
- the distance between two points with coordinates $\binom{2}{3}$ and $\binom{3}{5}$, or any other pair of points that are one unit apart in one coordinate direction and two in the other.

As is expected for algebraic structures defined on cartesian products, addition is defined **componentwise**:

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{pmatrix}$$

Given a real number c, we can scale elements of $\mathbb{R}^n$ by that number:

$$c \cdot \vec{x} = c \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} c \cdot x_1 \\ \vdots \\ c \cdot x_n \end{pmatrix}$$

The factor c is known as a ***scalar***, and the operator $\cdot : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n$ as ***scalar multiplication***. For the sake of readability, we differentiate scalars from vectors by writing a little arrow on top of a vector.

Arguably, it is these two operations - vector addition and scalar multiplication - that define linear algebra - next chapter, we will abstractly define ***vector spaces*** to be those structures where addition and scalar multiplication are well-behaved and the scalars come from a field, and ***modules*** to be their cousins whose scalars come from a ring.

In general, there is no well-behaved[2] vector multiplication that works for any n. Later on, we will prove this observation in the form of ***Frobenius' Theorem***, which roughly states:

[2] In this case, our criterion for multiplication being well-behaved is that multiplicative inverses should exist, i.e. that division should be possible.

Theorem 1.1.1: Frobenius' Theorem

The only dimensions in which an associative multiplication exists on $\mathbb{R}^n$ are $\mathbb{R}^1$, i.e. $\mathbb{R}$ itself, $\mathbb{R}^2$, where multiplication is given by treating two-dimensional vectors as complex numbers, and $\mathbb{R}^4$, where multiplication is given treating four-dimensional vectors as ***quaternions***.

We will take a closer look at the complex numbers $\mathbb{C} \cong \mathbb{R}^2$ and the quaternions $\mathbb{H} \cong \mathbb{R}^4$ in the following sections. In particular, we will see that quaternion multiplication isn't commutative, so if we require multiplication to be commutative, our options shrink even further, leaving just $\mathbb{R}$ and $\mathbb{C} \cong \mathbb{R}^2$.

1.2 The Inner Product: Angles and Lengths in $\mathbb{R}^n$

Another operation that is of central importance to euclidean geometry is the ***inner product***, also known as the ***dot product*** or ***scalar product***.

Definition 1.2.1: Euclidean Inner Product

Given two vectors $\vec{x}, \vec{y} \in \mathbb{R}^n$, their ***Euclidean inner product***, written $\langle \vec{x}, \vec{y} \rangle$ or simply $\vec{x} \cdot \vec{y}$, is the sum of the products of the components of $\vec{x}$ with the components of $\vec{y}$:

$$\langle \vec{x}, \vec{y} \rangle = \sum_{i=1}^{n} x_i \cdot y_i$$

Theorem 1.2.2: Properties of Inner Products

The Euclidean inner product has the following properties:

1. ***(Positive definiteness):*** For every $\vec{x} \in \mathbb{R}^n$, we have $\langle \vec{x}, \vec{x} \rangle = 0$ if and only if $\vec{x} = \vec{0}$.
2. ***(Linearity in the first argument):*** For every $\vec{x}, \vec{y}, \vec{z} \in \mathbb{R}^n$ and $a, b \in \mathbb{R}$, we have:

$$\langle ax + by, z \rangle = a \langle x, z \rangle + b \langle y, z \rangle$$

3. ***(Symmetry):*** For every $\vec{x}, \vec{y} \in \mathbb{R}^n$, we have:

$$\langle x, y \rangle = \langle y, x \rangle$$

Exercise 1.2.3. *Show that the Euclidean inner product fulfills the following version of the binomial formula:*

$$\langle \vec{x} + \vec{y}, \vec{x} + \vec{y} \rangle = \langle \vec{x}, \vec{x} \rangle + 2 \langle \vec{x}, \vec{y} \rangle + \langle \vec{y}, \vec{y} \rangle$$
$$\langle \vec{x} - \vec{y}, \vec{x} - \vec{y} \rangle = \langle \vec{x}, \vec{x} \rangle - 2 \langle \vec{x}, \vec{y} \rangle + \langle \vec{y}, \vec{y} \rangle$$

The importance of inner products comes from the fact they let us define ***lengths*** and ***angles***.

Definition 1.2.4: Euclidean Norm

The length of a vector $\vec{x} \in \mathbb{R}^n$, also known as its ***Euclidean norm***, written as $\|\vec{x}\|_2$, $\|\vec{x}\|$, or just $|x|$, is given by:

$$\|\vec{x}\|_2 = \sqrt{\langle \vec{x}, \vec{x} \rangle} = \sqrt{x_1^2 + \ldots + x_n^2}$$

Note that for $n = 1$, we simply get $\|x\|_2 = \sqrt{x^2} = |x|$, so any result concerning the Euclidean norm on $\mathbb{R}^n$ also holds for the absolute value function.

Theorem 1.2.5: Some properties of Norms

The Euclidean norm has the following properties:

1. ***(Non-negativity):*** For every $\vec{x} \in \mathbb{R}^n$, $\|\vec{x}\|_2 \geq 0$.
2. ***(Positive definiteness):*** For every $\vec{x} \in \mathbb{R}^n$, $\|\vec{x}\|_2 = 0$ if and only if $\vec{x} = \vec{0}$.
3. ***(Absolute homogeneity):*** For every $c \in \mathbb{R}$ and $\vec{x} \in \mathbb{R}^n$, we have

$$\|c \cdot \vec{x}\|_2 = |c| \cdot \|\vec{x}\|_2$$

The little subscript 2 in $\|\vec{x}\|_2$ differentiates the Euclidean norm from the many other norms we will meet over time - in general, for any $p \in \mathbb{N}_1$, $\|\vec{x}\|_p$ denotes the function

$$\|\vec{x}\|_p = \sqrt[p]{|x_1|^p + \ldots + |x_n|^p} \qquad \left(= \sqrt[p]{\sum_{i=1}^{n} |x_i|^p}\right),$$

known as the ***p-norm***. For now, we only really care about the Euclidean norm, but other values of p will start coming in handy once we start applying tools from analysis to vector spaces. The limiting case $p \to \infty$ is particularly useful: We will find that

$$\|\vec{x}\|_\infty = \max |x_i|.$$

Theorem 1.2.6: Cauchy-Schwartz Inequality

Let $\vec{x}, \vec{y} \in \mathbb{R}^n$. Then we have

$$|\langle \vec{x}, \vec{y} \rangle| \leq \|\vec{x}\|_2 \cdot \|\vec{y}\|_2$$

Proof.

1. Assume $\|\vec{x}\|_2 = 0$. Then we immediately get

$$\langle \vec{x}, \vec{y} \rangle = 0 = \|\vec{x}\|_2 \cdot \|\vec{y}\|_2.$$

2. Assume $\|\vec{x}\|_2 = \|\vec{y}\|_2 = 1$. Then we have $\|\vec{x}\|_2^2 + \|\vec{y}\|_2^2 = 2$, and thus:

$$\begin{aligned} 1 + \langle \vec{x}, \vec{y} \rangle &= \frac{1}{2}(\|\vec{x}\|_2^2 + \|\vec{y}\|_2^2) + \langle \vec{x}, \vec{y} \rangle \\ &= \frac{1}{2}(\|\vec{x}\|_2^2 + 2\langle \vec{x}, \vec{y} \rangle + \|\vec{y}\|_2^2) \\ &= \frac{1}{2}\|\vec{x} + \vec{y}\|_2^2 \\ &\geq 0 \end{aligned}$$

Thus we get $\langle \vec{x}, \vec{y} \rangle \geq -1$. Applying the same reasoning to $1 - \langle \vec{x}, \vec{y} \rangle$, we get

$$1 - \langle \vec{x}, \vec{y} \rangle = \frac{1}{2}\|\vec{x} - \vec{y}\|_2^2 \geq 0,$$

i.e. $\langle \vec{x}, \vec{y} \rangle \leq 1$. We thus have

$$|\langle \vec{x}, \vec{y} \rangle| \leq 1 = \|\vec{x}\|_2 \cdot \|\vec{y}\|_2,$$

as desired.

3. Now, let $\vec{x}, \vec{y} \in \mathbb{R}^n \setminus \left\{\vec{0}\right\}$ be arbitrary. Then $\frac{\vec{x}}{\|\vec{x}\|_2}$ and $\frac{\vec{y}}{\|\vec{y}\|_2}$ both have norm 1, letting us simply use our previous result:

$$\begin{aligned}
\left|\langle \vec{x}, \vec{y} \rangle\right| &= \left| \|\vec{x}\| \cdot \|\vec{y}\| \cdot \left\langle \frac{\vec{x}}{\|\vec{x}\|}, \frac{\vec{y}}{\|\vec{y}\|} \right\rangle \right| \\
&\leq \left| \|\vec{x}\| \cdot \|\vec{y}\| \cdot \left\| \frac{\vec{x}}{\|\vec{x}\|} \right\| \cdot \left\| \frac{\vec{y}}{\|\vec{y}\|} \right\| \right| \\
&= \|\vec{x}\| \cdot \|\vec{y}\|
\end{aligned}$$

□

Corollary 1.2.7: Triangle Inequality

For every $\vec{x}, \vec{y} \in \mathbb{R}^n$, we have

$$\|\vec{x} + \vec{y}\|_2 \leq \|\vec{x}\|_2 + \|\vec{y}\|_2$$

We had previously formalized the concept of an 'angle between two lines' via the complex exponential function, and then defined the trigonometric functions $\sin x$ and $\cos x$ as the complex and real parts of e^{ix}, respectively. If we take the cosine function as a given, then scalar products provide a convenient alternate way of defining angles, which works in any dimension:

Definition 1.2.8: Inner angle between two vectors

Given two vectors $\vec{x}, \vec{y} \in \mathbb{R}^n \setminus \left\{\vec{0}\right\}$, we define the ***inner angle*** between them to be:

$$\measuredangle(\vec{x}, \vec{y}) = \arccos\left(\frac{\langle \vec{x}, \vec{y} \rangle}{\|\vec{x}\|_2 \cdot \|\vec{y}\|_2} \right)$$

Note that Cauchy-Schwartz gives us $\frac{\langle \vec{x}, \vec{y} \rangle}{\|\vec{x}\|_2 \cdot \|\vec{y}\|_2} \in [-1, 1]$, showing that this expression actually is well-defined.

Exercise 1.2.9. *Verify that we have:*

1. $\measuredangle(\vec{x}, \vec{y}) = 0$ *if and only if there exists* $c \in \mathbb{R}_{>0}$ *such that* $\vec{x} = c\vec{y}$.
2. $\measuredangle(\vec{x}, \vec{y}) = \frac{\pi}{2}$ *if and only if* $\langle x, y \rangle = 0$.
3. $\measuredangle(\vec{x}, \vec{y}) = \pi$ *if and only if there exists* $c \in \mathbb{R}_{>0}$ *such that* $\vec{x} = -c\vec{y}$.

Notably, this suggests that our definition is sensible even in the case $n = 1$, since any pair of non-zero vectors fulfills either condition 1 or 2, implying that the angle between two vectors on the real number line is either 0 or π.

These inner angles match our earlier definition of angles in terms of complex numbers, as long as we are content with not being able to meaningfully talk about negative angles or angles greater than π anymore:

Exercise 1.2.10. *Earlier in this book, we defined angles in terms of complex numbers. Given a complex number $y = re^{i\varphi}$ we saw that the corresponding real vector is given by $\vec{y} = \binom{r\cos\varphi}{r\sin\varphi}$.*

Show that our two definitions match as long as $\varphi \in [0, \pi]$, i.e. that given a vector $\vec{x} = \binom{x}{0}$ lying on the x-axis, we have

$$\measuredangle(\vec{x}, \vec{y}) = \varphi$$

(Hint: Remember $\cos^2\varphi + \sin^2\varphi = 1$)

We have to be content with this restricted interval $[0, \pi]$, since the cosine function isn't injective on larger intervals, meaning the arccosine would no longer be well-defined.

More generally, any definition of an 'angle between two arbitrary vectors' will suffer from this - if we only specify two vectors, it's left ambiguous which side we should measure the angle on.

This problem can be solved by finding a way to take a tuple $(\vec{x}_1, \ldots, \vec{x}_n)$ of vectors and recognize whether it is 'positively ordered' or 'negatively ordered' - in $\mathbb{R}^2$, this would be asking whether the second vector lies 'clockwise' or 'counterclockwise' from the first, while in $\mathbb{R}^3$ the question is whether the vectors are 'left-handed' or 'right-handed'. A map that gives us this information is known as an ***orientation***. We will formalize this idea later, when we have introduced the necessary tools, those being vector spaces, their ***bases***, and ***determinant functions***.

In the complex case, we could dodge this issue because one of our vectors, call it $\vec{x}$, was always part of the x-axis, and thus we could obtain a canonical orientation by declaring that the angle between $\vec{x}$ and $\vec{y}$ was positive if $y_2 > 0$ and negative if $y_2 < 0$.

Exercise 1.2.11. *Prove the following theorems of classical Euclidean geometry:*

1. *The **Law of Cosines**:*

$$\|\vec{x} - \vec{y}\|_2^2 = \|\vec{x}\|_2^2 + \|\vec{y}\|^2 - 2\|\vec{x}\|_2\|\vec{y}\|_2 \cdot \cos\big(\measuredangle(\vec{x}, \vec{y})\big)$$

2. *The **Pythagorean Theorem**:*

$$\measuredangle(\vec{x}, \vec{y}) = \frac{\pi}{2} \implies \|\vec{x} - \vec{y}\|_2^2 = \|\vec{x}\|_2^2 + \|\vec{y}\|^2$$

1.3 The Cross Product

The ***cross product*** is an operation $\times : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}^3$, such that:

1. $\vec{u} \times \vec{v}$ is orthogonal to both $\vec{u}$ and $\vec{v}$,
2. the length of $\vec{u} \times \vec{v}$ is the area of the parallelogram spanned by $\vec{u}$ and $\vec{v}$:

$$\left\|\vec{u} \times \vec{v}\right\|_2 = \left\|\vec{u}\right\|_2 \left\|\vec{v}\right\|_2 \sin\left(\measuredangle(\vec{u}, \vec{v})\right)$$

We also declare that our cross product should be ***anticommutative:*** given two vectors, we want to be able to change the sign of the cross product by switching the order in which we multiply:

$$\vec{v} \times \vec{w} := -\vec{w} \times \vec{v}$$

Notice how this immediately implies

$$\begin{aligned} & \vec{v} \times \vec{v} = -(\vec{v} \times \vec{v}) \\ \implies & \vec{v} \times \vec{v} = 0 \end{aligned}$$

Finally, notice that our geometric intuition immediately tells us that our product should be ***linear*** and ***distributive over vector addition***.

We can now derive an explicit formula for this operation on our own. First, note that the standard basis

$$\vec{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \vec{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \vec{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix},$$

consists of orthogonal vectors of length one, so we can start by declaring

$$\begin{aligned} \vec{e}_1 \times \vec{e}_2 &:= \vec{e}_3 \\ \vec{e}_2 \times \vec{e}_3 &:= \vec{e}_1 \\ \vec{e}_3 \times \vec{e}_1 &:= \vec{e}_2 \end{aligned}$$

Note that once again, the direction in which these vectors point depends on a somewhat arbitrary choice of orientation - we could invert all of the directions and still end up with a valid definition.

These observations are already enough to calculate the cross product of two

arbitrary vectors: Using distributivity, we get:

$$
\begin{aligned}
\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \times \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} &= (v_1\vec{e}_1 + v_2\vec{e}_2 + v_3\vec{e}_3) \times (w_1\vec{e}_1 + w_2\vec{e}_2 + w_3\vec{e}_3) \\
&= v_1w_1(\vec{e}_1 \times \vec{e}_1) \\
&+ v_1w_2(\vec{e}_1 \times \vec{e}_2) \\
&+ v_1w_3(\vec{e}_1 \times \vec{e}_3) \\
\\
&+ v_2w_1(\vec{e}_2 \times \vec{e}_1) \\
&+ v_2w_2(\vec{e}_2 \times \vec{e}_2) \\
&+ v_2w_3(\vec{e}_2 \times \vec{e}_3) \\
\\
&+ v_3w_1(\vec{e}_3 \times \vec{e}_1) \\
&+ v_3w_2(\vec{e}_3 \times \vec{e}_2) \\
&+ v_3w_3(\vec{e}_3 \times \vec{e}_3)
\end{aligned}
$$

To which we can apply anticommutativity and our formulas for multiplying the basis vectors to get:

$$
\begin{aligned}
&\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \times \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} \\
\\
&= v_1w_1 \cdot \vec{0} \\
&+ v_1w_2 \cdot \vec{e}_3 \\
&+ v_1w_3 \cdot (-\vec{e}_2) \\
\\
&+ v_2w_1 \cdot (-\vec{e}_3) \\
&+ v_2w_2 \cdot \vec{0} \\
&+ v_2w_3 \cdot \vec{e}_1 \\
\\
&+ v_3w_1 \cdot \vec{e}_2 \\
&+ v_3w_2 \cdot (-\vec{e}_1) \\
&+ v_3w_3 \cdot \vec{0} \\
\\
&= \begin{pmatrix} v_2w_3 - v_3w_2 \\ v_3w_1 - w_1v_3 \\ v_1w_2 - v_2w_1 \end{pmatrix}
\end{aligned}
$$

Corollary 1.3.1: Cross Product

The cross product $\vec{v} \times \vec{w}$ of two vectors $\vec{v}, \vec{w} \in \mathbb{R}^3$ returns a vector whose direction is orthogonal to both $\vec{v}$ and $\vec{w}$ and whose length equals the area of the parallelogral spanned between them. Given coordinate representations of $\vec{v}$ and $\vec{w}$, the cross product can be calculated via the following formula:

$$\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \times \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} = \begin{pmatrix} v_2 w_3 - v_3 w_2 \\ v_3 w_1 - w_1 v_3 \\ v_1 w_2 - v_2 w_1 \end{pmatrix}$$

Proposition 1.3.2. *We have*

$$\left\langle \vec{u} \times \vec{v}, \vec{w} \times \vec{t} \right\rangle = \left\langle \vec{u}, \vec{w} \right\rangle \left\langle \vec{v}, \vec{t} \right\rangle - \left\langle \vec{u}, \vec{t} \right\rangle \left\langle \vec{v}, \vec{w} \right\rangle .$$

Proposition 1.3.3. *We have*

$$\left\langle \vec{u} \times \vec{v}, \vec{w} \right\rangle = \left\langle \vec{v} \times \vec{w}, \vec{u} \right\rangle = \left\langle \vec{w} \times \vec{u}, \vec{v} \right\rangle$$

This expression is known as the ***box product*** *of $\vec{u}, \vec{v}, \vec{w}$. Its absolute value corresponds to the volume of the parallelepiped spanned by the three vectors.*

Proposition 1.3.4. *We have*

$$(\vec{u} \times \vec{v}) \times \vec{w} = \langle u, w \rangle \cdot v - \langle v, w \rangle \cdot u = w \times (v \times u).$$

This result is known as ***Grassmann's Identity****.*

Proposition 1.3.5. *We have*

$$(\vec{u} \times \vec{v}) \times \vec{w} + (\vec{v} \times \vec{w}) \times \vec{u} + (\vec{w} \times \vec{u}) \times \vec{v} = 0$$

This result is known as ***Jacobi's Identity.***

1.4 The Quaternions $\mathbb{H}$

Definition 1.4.1: Quaternions

The Quaternions $\mathbb{H}$ consist of the set $\mathbb{R}^4$, with the following special notation:

$$\begin{aligned}\vec{1} &:= (1,0,0,0)\\ i &:= (0,1,0,0)\\ j &:= (0,0,1,0)\\ k &:= (0,0,0,1)\end{aligned}$$

Such that in addition to scalar multiplication and vector addition, which work as usual, we have an additional vector multiplication operation $\cdot : \mathbb{R}^4 \times \mathbb{R}^4 \to \mathbb{R}^4$ that behaves as follows:

1. $\vec{1}$ is a neutral element of vector multiplication:

$$\begin{aligned}&\forall x \in \mathbb{H} :\\ &\vec{1} \cdot x = x \cdot \vec{1} = x\end{aligned}$$

2. Vector multiplication is associative:

$$\vec{x}(\vec{y}\vec{z}) = (\vec{x}\vec{y})\vec{z}$$

3. Multiplicative inverses exist:

$$\begin{aligned}&\forall \vec{x} \in \mathbb{H}_{\neq 0} :\\ &\exists \vec{x}^{-1} \in \mathbb{H} :\\ &\vec{x} \cdot \vec{x}^{-1} = \vec{x}^{-1} \cdot \vec{x} = \vec{1}\end{aligned}$$

4. Vector multiplication distributes over vector addition, from both sides:

$$\begin{aligned}&\forall \vec{x}, \vec{y}, \vec{z} \in \mathbb{H} :\\ &\vec{x} \cdot (\vec{y} + \vec{z}) = \vec{x}\vec{y} + \vec{x}\vec{z}\\ &(\vec{x} + \vec{y}) \cdot \vec{z} = \vec{x}\vec{z} + \vec{y}\vec{z}\end{aligned}$$

5. Vector multiplication is compatible with scalar multiplication:

$$\begin{aligned}&\forall a, b \in \mathbb{R}, \vec{x}, \vec{y} \in \mathbb{H} :\\ &a\vec{x} \cdot b\vec{y} = (ab)(\vec{x}\vec{y})\end{aligned}$$

6. The unit vectors fulfill the ***Quaternion multiplication formula***:

$$i^2 = j^2 = k^2 = ijk = -\vec{1}$$

Apart from the Quaternion multiplication formula, which is the defining feature of the Quaternions, all of our other demands together state that $\mathbb{H}$ should be an ***associative real division algebra over*** $\mathbb{R}^4$.

One very important consequence of this definition is that vectors of the form $\vec{a} := (a, 0, 0, 0)$ behave exactly like real numbers - let $\vec{x} \in \mathbb{H}$, then we have:

$$\vec{a}\vec{x} = (a \cdot \vec{1}) \cdot \vec{x} = a \cdot (\vec{1} \cdot \vec{x}) = a\vec{x}$$

For this reason, it is common to write $\vec{1}$ simply as 1, and to then proceed as we did for the complex numbers and write:

$$(a, b, c, d) = a + bi + cj + dk$$

We can now already get most of the way to deriving multiplication formula for quaternions, by simply applying distributivity:

$$\begin{aligned}
&(v_1 + v_2 i + v_3 j + v_4 k) \cdot (u_1 + u_2 i + u_3 j + u_4 k) \\
&\qquad = v_1 u_1 + (v_1 u_2) i + (v_1 u_3) j + (v_1 u_4) k \\
&\qquad + (v_2 u_1) i + (v_2 u_2) i^2 + (v_2 u_3) ij + (v_2 u_4) ik \\
&\qquad + (v_3 u_1) j + (v_3 u_2) ji + (v_3 u_3) j^2 + (v_3 u_4) jk \\
&\qquad + (v_4 u_1) k + (v_4 u_2) ki + (v_4 u_3) kj + (v_4 u_4) k^2
\end{aligned}$$

Now, using $i^2 + j^2 + k^2 = ijk = -1$, we can simplify slightly:

$$\begin{aligned}
&(v_1 + v_2 i + v_3 j + v_4 k) \cdot (u_1 + u_2 i + u_3 j + u_4 k) \\
&\qquad = v_1 u_1 + (v_1 u_2) i + (v_1 u_3) j + (v_1 u_4) k \\
&\qquad + (v_2 u_1) i + (v_2 u_2) i^2 + (v_2 u_3) ij + (v_2 u_4) ik \\
&\qquad + (v_3 u_1) j + (v_3 u_2) ji + (v_3 u_3) j^2 + (v_3 u_4) jk \\
&\qquad + (v_4 u_1) k + (v_4 u_2) ki + (v_4 u_3) kj + (v_4 u_4) k^2 \\
&\qquad = v_1 u_1 + (v_1 u_2) i + (v_1 u_3) j + (v_1 u_4) k \\
&\qquad + (v_2 u_1) i - (v_2 u_2) + (v_2 u_3) ij + (v_2 u_4) ik \\
&\qquad + (v_3 u_1) j + (v_3 u_2) ji - (v_3 u_3) + (v_3 u_4) jk \\
&\qquad + (v_4 u_1) k + (v_4 u_2) ki + (v_4 u_3) kj - (v_4 u_4)
\end{aligned}$$

All that's left is to figure out what the products of i, j and k should be.

We can derive ij and ji as follows:

1. Start with $ijk = -1$.
2. Multiply k from the right to get $ijk^2 = -k$, i.e. $-ij = -k$, i.e. $ij = k$.
3. Multiply ij from the left to get $ijij = ijk = -1$.
4. Multiply i from the left and j from the right to get $i(ijij)j = -ij$.
5. Simplify the left side to get $i(ijij)j = (ii)(ji)(jj) = ji$, i.e:

$$ij = -ji = k$$

Exercise 1.4.2. *Show that we also have $jk = -kj = i$ and $ki = -ik = j$.*

This means that Quaternion multiplication isn't commutative!

Our final multiplication table for $1, i, j$ and k is thus:

$\cdot$	1	i	j	k
1	1	i	j	k
i	i	-1	k	$-j$
j	j	$-k$	-1	i
k	k	j	$-i$	-1

Exercise 1.4.3. *Verify that this multiplication table doesn't violate associativity.*

Entering these new results into our partial multiplication formula, we get:

$$\begin{aligned}
&(v_1 + v_2 i + v_3 j + v_4 k) \cdot (u_1 + u_2 i + u_3 j + u_4 k) \\
&\qquad = v_1 u_1 + (v_1 u_2) i + (v_1 u_3) j + (v_1 u_4) k \\
&\qquad + (v_2 u_1) i - (v_2 u_2) + (v_2 u_3) ij + (v_2 u_4) ik \\
&\qquad + (v_3 u_1) j + (v_3 u_2) ji - (v_3 u_3) + (v_3 u_4) jk \\
&\qquad + (v_4 u_1) k + (v_4 u_2) ki + (v_4 u_3) kj - (v_4 u_4) \\
&\qquad = v_1 u_1 + (v_1 u_2) i + (v_1 u_3) j + (v_1 u_4) k \\
&\qquad + (v_2 u_1) i - (v_2 u_2) + (v_2 u_3) k - (v_2 u_4) j \\
&\qquad + (v_3 u_1) j - (v_3 u_2) k - (v_3 u_3) + (v_3 u_4) i \\
&\qquad + (v_4 u_1) k + (v_4 u_2) j - (v_4 u_3) i - (v_4 u_4)
\end{aligned}$$

which we can finally simplify to get:

$$\begin{aligned}
&(v_1 + v_2 i + v_3 j + v_4 k) \cdot (u_1 + u_2 i + u_3 j + u_4 k) \\
&\qquad = \ (v_1 u_1 - v_2 u_2 - v_3 u_3 - v_4 u_4) \\
&\qquad + (v_1 u_2 + v_2 u_1 + v_3 u_4 - v_4 u_3) i \\
&\qquad + (v_1 u_3 - v_2 u_4 + v_3 u_1 + v_4 u_2) j \\
&\qquad + (v_1 u_4 + v_2 u_3 - v_3 u_2 + v_4 u_1) k
\end{aligned}$$

Now, this is obviously a long and unreasonably complicated formula - is there any way to think of it differently? The complex numbers had a very nice and intuitive geometric definition, maybe the quaternions do too?

Well, the key trick here is to split the underlying set $\mathbb{R}^4$ into $\mathbb{R} \times \mathbb{R}^3$, thinking of each Quaternion as consisting of a ***real part*** $a \in \mathbb{R}$ and an ***imaginary part*** $\vec{v} \in \mathbb{R}^3$. Our multiplication formula now reads:

$$\begin{aligned}
&(a, \vec{v}) \cdot (b, \vec{u}) \\
&= \ (ab - v_1 u_1 - v_2 u_2 - v_3 u_3) \\
&+ (a u_1 + b v_1 + v_2 u_3 - v_3 u_2) i \\
&+ (a u_2 - v_1 u_3 + b v_2 + v_3 u_1) j \\
&+ (a u_3 + v_1 u_2 - v_2 u_1 + b v_3) k
\end{aligned}$$

which you might notice consists exclusively of familiar vector operations:

1. $ab - v_1u_1 - v_2u_2 - v_3u_3 = ab - \left\langle \vec{v}, \vec{u} \right\rangle$
2. $\begin{pmatrix} au_1 \\ au_2 \\ au_3 \end{pmatrix} = a\vec{u}, \begin{pmatrix} bv_1 \\ bv_2 \\ bv_3 \end{pmatrix} = b\vec{v}$
3. $\begin{pmatrix} v_2u_3 - v_3u_2 \\ v_1u_3 - v_3u_1 \\ v_1u_2 - v_2u_1 \end{pmatrix} = \vec{v} \times \vec{u}$

Putting everything together, we can now write out our multiplication formula in a much nicer form:

$$(a, \vec{v}) \cdot (b, \vec{u}) = (ab - \left\langle \vec{v}, \vec{u} \right\rangle, a\vec{u} + b\vec{v} + \vec{v} \times \vec{u}),$$

Exercise 1.4.4. *Verify that quaternion multiplication is associative and distributes over vector addition.*

Exercise 1.4.5. *Let $p = (a, \vec{v}) \in \mathbb{H}$. Then $pq = qp$ holds for all $q \in \mathbb{H}$ if and only if $\vec{v} \neq 0$.*

Similar to complex numbers, we can define ***quaternion conjugation***:

Definition 1.4.6: Quaternion Conjugation

Just like for $\mathbb{C}$, we define the ***conjugate*** of a quaternion $(a, \vec{v})$ to be the quaternion

$$\overline{(a, \vec{v})} = (a, -\vec{v})$$

An immediate application of quaternion conjugation is finding a formula for the inverse of a quaternion, which ones again ends up working out very similarly to the complex case - We have

$$\begin{aligned} \overline{(a, \vec{v})} \cdot (a, \vec{v}) &= (a, -\vec{v}) \cdot (a, \vec{v}) \\ &= (a^2 - \left\langle -\vec{v}, \vec{v} \right\rangle, a\vec{v} - a\vec{v} + (-\vec{v} \times \vec{v})) \\ &= \left(a^2 + \left\|\vec{v}\right\|^2, \vec{0}\right) \in \mathbb{R} \end{aligned}$$

And thus we can simply divide by this value to get our inverse formula:

Corollary 1.4.7: Inverse Quaternion

Let $(a, \vec{v}) \in \mathbb{H}_{\neq 0}$. Then $(a, \vec{v})$ has a multiplicative inverse, which is given by:

$$(a, \vec{v})^{-1} = \frac{(a, -\vec{v})}{a^2 + \|v\|^2}.$$

Definition 1.4.8. Let $q = (a, \vec{v}) \in \mathbb{H}$. Then

$$|q| := \sqrt{\overline{q} \cdot q} = a^2 + v_1^2 + v_2^2 + v_3^2 = \|q\|_2$$

Note how much this once again resembles both the real case $|x| = \sqrt{x^2} = \|x\|_2$ and the complex case $|z| = \sqrt{\overline{z}z} = \|z\|_2$.

1.5 Quaternions and $\mathbb{R}^3$

Theorem 1.5.1: Quaternions encode rotations

Let $\|\vec{v}\| = 1$ and

$$q = (\cos\varphi, \vec{v}\sin\varphi) \in \mathbb{H}$$

Then $qw\overline{q} \in \mathbb{R}^3$ is the vector obtained by rotating w around the axis $\{\lambda\vec{v} : \lambda \in \mathbb{R}\}$ by the angle 2φ.

Since 2π corresponds to a full rotation, φ and $\varphi + \pi$ describe the same rotation. Therefore, there are two different quaternions q and $-q$ associated with each rotation - in topological terms, we say that the quaternions form a ***double cover*** of the space of all rotations.

This phenomenon is actually closely related to the theory of ***spin*** in quantum physics. All known quantum particles that constitute ordinary matter, including protons, neutrons, electrons, neutrinos, and quarks, are ***spin-$\frac{1}{2}$***, meaning that they exhibit a curious behavior where a rotation of 360° leaves them in a different state than they were before, from which they return to their original state only after being rotated by another 360° around the same axis. This lets us mathematically model particles as quaternions, such that a 360° rotation transforms a particle from a state $q \in \mathbb{H}$ to the "rotationally equivalent" state $-q \in \mathbb{H}$.

Corollary 1.5.2. *For every isometry $F : \mathbb{R}^3 \to \mathbb{R}^3$, there exist $u \in \mathbb{R}^3$ and $q \in \mathbb{H}$ such that we have either*

1. $F(w) = u + qw\overline{q}$, *if F preserves orientation, or*
2. $F(w) = u + q\overline{wq}$, *if F doesn't preserve orientation.*

This representation turns out to be a lot more computationally efficient than other representations of the same rotation. Other, more intuitively "trivial" representations of rotation also run into issues such as ***gimbal lock***, where they may get "stuck" in certain orientations.

This makes quaternions surprisingly practical in many applied contexts, including game development and animation[1], robotics, and three-dimensional image processing.

[1]For an introduction to quaternions in game development, I highly recommend this excellent talk by Freya Holmer.

Chapter 2

Modules and Vector Spaces

2.1 Linear Maps

2.2 Modules and Vector Spaces

2.3 Subspaces and Quotient Spaces

2.4 Matrices

2.5 Dimension

2.6 Systems of Linear Equations

2.7 The Method of Least Squares

Chapter 3

Determinants

3.1 Defining Volume

3.2 Determinants of Endomorphisms

3.3 Orientation

Chapter 4

Eigenvectors and Eigenvalues

Definition 4.0.1: Eigenvalues and Eigenvectors

Let k be a field, $F : V \to V$ an Endomorphism, and $\lambda \in k$. Then the ***λ-eigenspace*** of F is the set

$$\mathrm{Eig}(F, \lambda) := \left\{\vec{v} \in V : F(\vec{v}) = \lambda\vec{v}\right\} = \left\{\vec{v} \in V : (F - \lambda \mathrm{id}_V)(\vec{v}) = 0\right\} = \ker(F - \lambda \mathrm{id}_V)$$

If $F(\vec{v}) = \lambda\vec{v}$, with $\vec{v} \neq \vec{0}$, we say that $\vec{v}$ is an ***eigenvector*** of F with ***eigenvalue*** λ.

Note that we exclude $\vec{0}$ from being an eigenvector, since $F(\vec{0}) = \lambda\vec{0}$ trivially holds for every endomorphism. Thus every λ would be an eigenvalue, which would make the definition pointless. We do, however, allow 0 as an eigenvalue - the 0-eigenspace of F is simply its kernel. Sometimes it ends up being useful to rephrase our definition of eigenspaces in terms of kernels:

$$\begin{aligned}\mathrm{Eig}(F, \lambda) &:= \left\{\vec{v} \in V : F(\vec{v}) = \lambda\vec{v}\right\} \\ &= \left\{\vec{v} \in V : (F - \lambda \mathrm{id}_V)(\vec{v}) = 0\right\} \\ &= \ker(F - \lambda \mathrm{id}_V)\end{aligned}$$

This shows that $\mathrm{Eig}(F, \lambda)$ is a subspace of V - by definition, these subspaces must then be the maximal subspaces that are closed under application of F, i.e. $F(\mathrm{Eig}(F, \lambda)) = \mathrm{Eig}(F, \lambda)$.

Exercise 4.0.2. *In an informal, intuition-based way:*

1. *Find the real eigenvalues and eigenspaces of a reflection of $\mathbb{R}^2$ along an axis $\left\{\lambda\vec{v} : \lambda \in \mathbb{R}\right\}$*
2. *Find a rotation of $\mathbb{R}^2$ that has no real eigenvalues. Find two different rotations of $\mathbb{R}^2$ that do have real eigenvalues. Show that any rotation of $\mathbb{R}^2$ has complex eigenvalues.*
3. *Find the eigenvectors of the derivative map $\frac{d}{dx}$*

Our characterisation $\mathrm{Eig}(F, \lambda) = \ker(F - \lambda \mathrm{id}_V)$ means that, if F is finite-dimensional, we can find all eigenvectors of F with a <u>given</u> eigenvalue λ by solving $(F - \lambda \mathrm{id}_V)\vec{v} = 0$ as a systen of linear equations. Finding the correct

eigenvalues ends up being more tricky - in the section after this one, we will work towards proving the following key result:

Proposition 4.0.3. *The eigenvalues of F are given by the zeroes of the* ***characteristic polynomial*** $\chi_F(\lambda) := \det(F - \lambda \mathrm{id}_V)$.

However, before we do so, we want to familiarize ourselves with some of the more basic properties of eigenvectors and eigenvalues.

Theorem 4.0.4. *Let V be a vector space over a field k. Let $B = (b_1, \ldots, b_n)$ be a basis of V, and let F be an endomorphism represented in B by a matrix A. Then:*

1. *A basis vector b_i is an eigenvector of F iff. the j-th column of A has the form λe_j.*
2. *F is diagonalizable iff. there exists a basis of V consisting of eigenvectors of F.*

Theorem 4.0.5. *Let V be a vector space over a field k. Let $F : V \to V$ be an endomorphism. Then eigenvectors of F with different eigenvalues are linearly independent.*

Corollary 4.0.6. *Let V be an n-dimensional vector space over a field k. Let $F : V \to V$ be an endomorphism. Then:*

1. *F has no more than n distinct eigenvalues,*
2. *If $(\lambda_i)_{i \in I}$ are mutually distinct eigenvalues of F, the sum*

$$\sum_{i \in I} \mathrm{Eig}(F, \lambda_i) \subseteq V$$

   is direct.

3. *F is diagonalizable if and only if V is a direct sum of eigenspaces of F.*

4.1 Characteristic Polynomials

4.2 The Cayley-Hamilton Theorem

4.3 Minimal Polynomials

Chapter 5

The Jordan Normal Form

Chapter 6

Inner Product Spaces

Part IV

Point-Set Topology

Chapter 1

Topological Spaces

A very common axiom is the Hausdorff separation axiom, which says that any two distinct points can be housed orf from each other by disjoint open sets.

"Topology via Logic",
Steven Vickers